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Reach Remote: An Unrivalled Disruptive Solution for Subsea WROV Operations

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Abstract

Reach Remote stands as a transformative technology in the realm of subsea operations, presenting a solution that is not only disruptive but truly capable of redefining the industry's approach to remote, uncrewed interventions. This manuscript elucidates the capabilities and significance of Reach Remote, Norway's pioneering 24-meter uncrewed surface vessel (USV) fleet. By seamlessly integrating advanced payload options, that include a Work Class Electric Remotely Operated Vehicle (WROV), Reach Remote can perform subsea operations globally without personnel at sea. This paper highlights the objectives, core technological advancements, operational processes, and the broader impact on sustainability, the energy sector, and defence applications that USVs can unlock.

Introduction

The offshore subsea sector has long been defined by its reliance on crewed vessels, complex logistics, and relatively significant operational costs. Conventional subsea vessel operations, often with work-class ROVs and subsea cranes, are resource-intensive and contribute heavily to marine emissions. The reality for many subsea vessel operators is that the full capability of the vessel, its equipment, and crew are often only used to full capacity for a small percentage of the time.

To put that another way, 80% of the time a DP2 subsea vessel is used to only a fraction and often less, of its full potential capability.

"80% of the capability of a DP2 subsea vessel is wasted"

With the increased focus across the globe on sustainable operations, it's hard to ignore the potential benefit to everyone from looking to improve that metric.

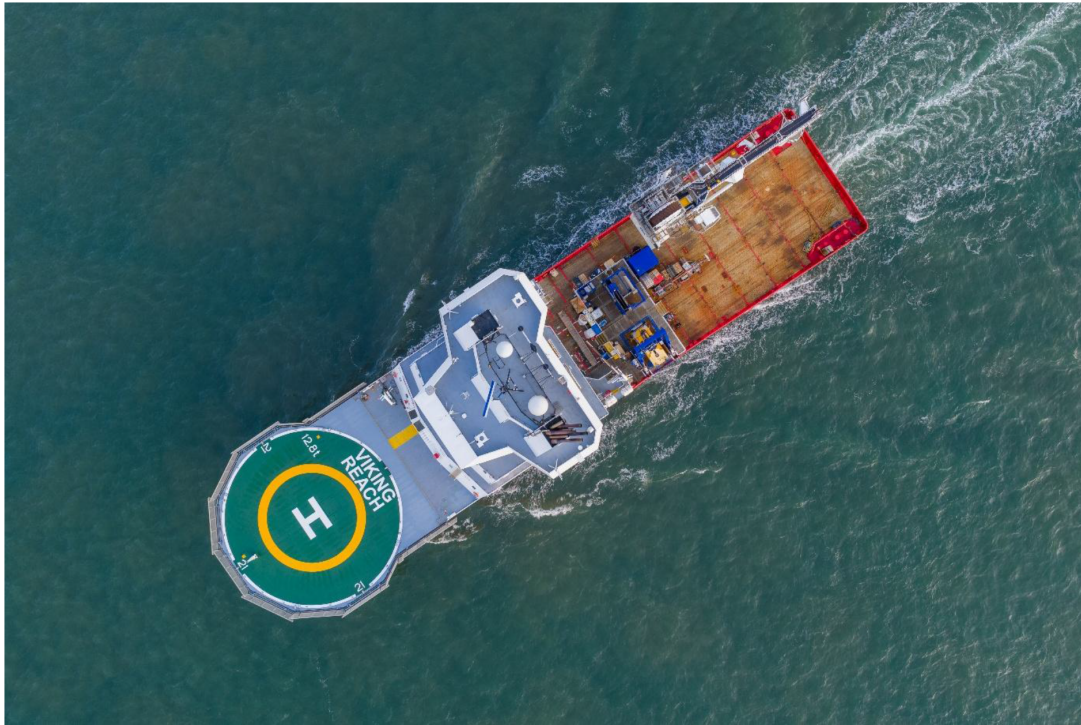


Figure 1—A typical DP2 subsea vessel [Image courtesy of Reach Subsea]

The arrival of Uncrewed Surface Vessels (USVs) signals a paradigm shift in the way maritime operations are conducted. Across all sectors; logistics, energy and defence. USV's are by no means new, but they are still relatively immature in the various markets in which they are beginning to operate - which presents an ever-evolving barrage of challenges, and opportunities, along the way.

The advent of Reach Remote marks a pivotal departure from convention, introducing fully uncrewed, remotely supervised operations with substantial reductions in both risk and cost. This paper presents a comprehensive overview of Reach Remote's capabilities, operational achievements, and its role in shaping the future of the offshore energy and naval defence sectors.

Objectives and Scope

USVs have the potential to disrupt the maritime sector, in an incredibly positive way. USVs, such as Reach Remote, are not merely an incremental improvement; it represents a paradigm shift, one that promises to dramatically enhance safety, efficiency, and sustainability.

- USVs do not mean less opportunities for people to work within the maritime sector. One of the biggest barriers to 'acceptance' of USVs is the human attitude towards change, and the perception that they "remove jobs". This is far from the case.
- Demonstrate the technical architecture and operational model behind Reach Remote.
- Explain how uncrewed, over-the-horizon maritime operations are enabled and supervised entirely from onshore control centres.
- Highlight the environmental and economic benefits derived from a 90% reduction in fuel consumption.
- Illustrate the broader implications for surveillance, defence, and energy infrastructure monitoring—without exposing personnel to offshore hazards.
- Situate Reach Remote within the context of global sustainability and digitalisation initiatives.

Methods, Procedures, and Processes

The Role of USVs in Modern Maritime Operations

Uncrewed Surface Vessels (USVs) represent a significant evolution in maritime capability. Operating without onboard crew, they can be deployed for a wide range of missions—from hydrographic survey and environmental monitoring to defence surveillance and subsea intervention.

Key characteristics of modern USVs include:

- **Remote or autonomous operation:** Controlled from shore-based facilities or pre-programmed for autonomous navigation.
 - The difference between remote and autonomous operations are often misunderstood and play a significant factor in the change management process during the introduction of USVs.
- **Mission flexibility:** Modular payloads allow rapid reconfiguration for different sectors, including offshore energy, defence, logistics, and scientific research.
- **Operational efficiency:** Smaller platforms with lower propulsion requirements drastically reduce fuel consumption and emissions.
- **Extended endurance:** Many designs can remain on station for weeks or months without resupply.
- **Safety enhancement:** By removing crew from high-risk offshore environments, USVs eliminate human exposure to hazardous conditions.

The USV sector is diverse, encompassing vessels from under 10 metres to over 40 metres in length. Technologies are advancing rapidly in hull design, propulsion efficiency, autonomous navigation, and secure communications. However, challenges remain in regulatory acceptance, standardisation, and integration with existing maritime infrastructure.



Figure 2—Sea Hunter USV - US Navy [https://en.wikipedia.org/wiki/Sea_Hunter]

Uncrewed Vessel Design and Capabilities

Reach Remote 1 is the flagship of Norway's first fleet of uncrewed 24-meter surface vessels. Engineered and built by Kongsberg, each vessel is equipped with:

- Hull-mounted survey sensors for real-time bathymetric and geophysical data acquisition.
- A Work Class Electric ROV (WROV), capable of full-spectrum intervention and inspection tasks, including manipulator deployment, pipeline inspection, and asset integrity assessments.
- Redundant propulsion and dynamic positioning systems, enabling stable operations in harsh environments and significant sea states.
- Robust satcom and redundant data channels, guaranteeing persistent high-bandwidth connectivity to shore-based command centres.
- A sustainable energy profile, with efficient propulsion and optimised hydrodynamics for minimum fuel consumption.
- The ability to carry anti-submarine warfare sensors and tooling packages in support of naval defence and surveillance applications.



Figure 3—Reach Remote 1 [Image courtesy of Reach Subsea]

Uncrewed Surface Vessels (USVs) should not be viewed as standalone assets but rather as integral components within a broader "system of systems." While the vessel itself, and the hardware installed on board, often attracts the most attention, it represents only one part of the overall operational framework. For a USV to function effectively and deliver on its intended mission, several interconnected elements must work together seamlessly.

At a high level, the USV ecosystem can be understood as comprising three key pillars:

COMPONENTS OF THE USV SYSTEM



**USV HULL
AND HARDWARE
AT SEA**



**CONTROL/
OVERSIGHT
FACILITY SYSTEM
ONSHORE**



**COMMUNICATION
SYSTEM AND
DATA FLOW
ARCHITECTURE**

- **The USV hull and onboard hardware at sea:** This includes the vessel platform itself, propulsion and power systems, onboard control and automation systems, and any mission-specific payloads (e.g., sensors, survey equipment, or robotic intervention tools). While the vessel is the most visible element, its effectiveness is dependent on integration with the wider system.
- **Control and oversight facilities onshore:** Operations require a shore-based facility that provides supervisory control, monitoring, and decision support. These facilities may be centralised in one location or distributed across multiple control centres. They allow human operators and specialists to oversee mission progress, ensure regulatory compliance, and intervene if necessary.
- **Communication systems and data flow architecture:** A robust and resilient communication link between the USV and the control facility is essential. This architecture enables real-time command and control, situational awareness, and continuous data transfer from vessel sensors back to shore. It must also account for redundancy, cybersecurity, and bandwidth management to ensure uninterrupted operations.

When evaluating the impact or effectiveness of a USV solution, it is therefore not enough to assess the vessel in isolation. The true measure of success lies in how well the entire system integrates with the mission objectives and operational environment. This means considering not only the technical performance of the vessel but also the reliability of communications, the effectiveness of shore-based oversight, and the ability of the full system to deliver actionable results in line with project and/or mission goals.

Command, Control, and Communication

The vessels are operated entirely from onshore control centres, leveraging intelligent software and advanced communication protocols. Key features include:

- Remote piloting and supervision by skilled operators using real-time video, sonar, and telemetry.
- Automated health monitoring and predictive maintenance powered by machine learning algorithms.

- Secure cyber-resilient communication pathways for both command-and-control and client data streaming.
- Flexible integration that allows end-users and clients to interface directly with the vessel's systems and sensors in real time, regardless of geographic location.

A key feature of the control philosophy behind many USV systems is the fact that onshore control facilities can be combined virtually to enable different locations and teams of experts to provide assistance to the operation at sea, on demand. This allows for immediate full team situational awareness and faster decision making. Ultimately leading to a more efficient campaign at sea.

Endurance and Operational Envelope

Reach Remote's design enables continuous offshore operation for over 30 days without the need for preventative maintenance or fuel. The vessel is capable of performing launch and recovery of the WROV in sea states up to 3.5m significant wave height, expanding the operational weather window beyond that of many current USVs in the market.

With an operating window similar to many existing subsea vessels, a significant advantage of USV systems over conventional DP2 subsea vessels is the fuel consumption. As mentioned at the beginning, 80% of the time DP2 subsea vessels are used they are underutilised. During that period of underutilisation, the conventional DP2 subsea vessel still consumes anything between 6 and 18 tonnes of fuel per day.

USVs can go out and do the same work scope whilst only consuming a fraction of the fuel.

As an example, Reach Remote 1 in ROV operations mode at sea consumes 0.6 - 0.9 tonnes of fuel per day. The same operation performed by the DP2 subsea vessel Viking Reach would consume circa 10 tonnes of fuel per day.

Results, Observations, and Conclusions

Operational Achievements

Since its launch, Reach Remote has undertaken multiple projects with global energy operators, including complex subsea inspections, pipeline surveys, and intervention campaigns. In each instance, the vessel has demonstrated:

- A 90% reduction in fuel consumption compared to a conventional DP2 ROV support vessel, contributing to significant emission reductions aligned with international sustainability goals.
- Zero personnel offshore, eliminating exposure to maritime risk and reducing insurance and logistics costs.
- Uninterrupted operations in demanding environments, validated by independent third-party audits and client feedback.

Case Study: Joint Industry Pilot Project - A pathway to credible USV operations

Uncrewed Surface Vessels (USVs) present a significant opportunity for sustainable operations at sea, across energy, logistics, defence, and other sectors. However, a key barrier to commercial adoption is market perception: many stakeholders see increased remote and autonomous operations, combined with crews based onshore rather than offshore, as a risk.

Potential clients often look to "de-risk" the introduction of new technology, sometimes without fully understanding the actual risks involved. Many require proof of concept or bespoke trials before committing, which can be costly and difficult for operators without commercial scale.

With the arrival of Reach Remote, Reach Subsea faced the challenge of securing industry validation quickly enough to enable commercial operations, while demonstrating the system's full capabilities to diverse clients.

To address this, Reach Subsea launched a Joint Industry Pilot (JIP). The objective was straightforward: introduce the Reach Remote platform and associated systems to a varied client base, while showcasing immediate, mission-relevant capabilities. Crucially, participation required both active engagement and financial commitment from stakeholders, as the scale of Reach Remote does not allow for multiple zero-cost trials.

The JIP brought together Reach Subsea with the following partners:

- Equinor
- Total Energies
- Harbour Energy
- Woodside Energy
- Shell
- Gassco
- Royal Australian Navy

Together, the partners developed a pilot program designed to deliver value to each stakeholder, demonstrate practical use cases, and build a deeper understanding of future opportunities with USVs.

The JIP centred on the USV Reach Remote 1, supported by:

- Control centres in Haugesund, Horten, and Oslo (Norway)
- A communications and data architecture enabling real-time system connection via Reach Horizon

Reach Horizon provided mission-critical oversight, allowing operators to command, control, and monitor the USV and its payloads in real time.

The JIP covered a wide range of offshore operations, including:

- Pipeline inspection on live assets in operational environments
- Structural inspection, including cathodic protection measurements
- gWatch reservoir modelling measurements
- USV hull and ROV seabed mapping
- Environmental surveys via ROV
- Guard vessel duties
- Subsea intervention and torque tool demonstrations

Operations took place offshore at Troll, Gullfaks, and Snorre in the North Sea, and at Asgard in the Norwegian Sea.

The JIP ran over 38 days, delivering excellent vessel and system performance:

- **100% scope completion:** Every task successfully demonstrated, often exceeding client expectations.
- **97% system uptime:** Only 4% of total time lost to weather delays.
- **Strong ROV performance:** Launch and recovery operations proved highly effective, even in challenging sea states—a critical factor for system uptime.

- **Reliable connectivity:** Minor carrier issues identified and captured in lessons learned. The Reach Horizon platform was central to real-time data access and system oversight.

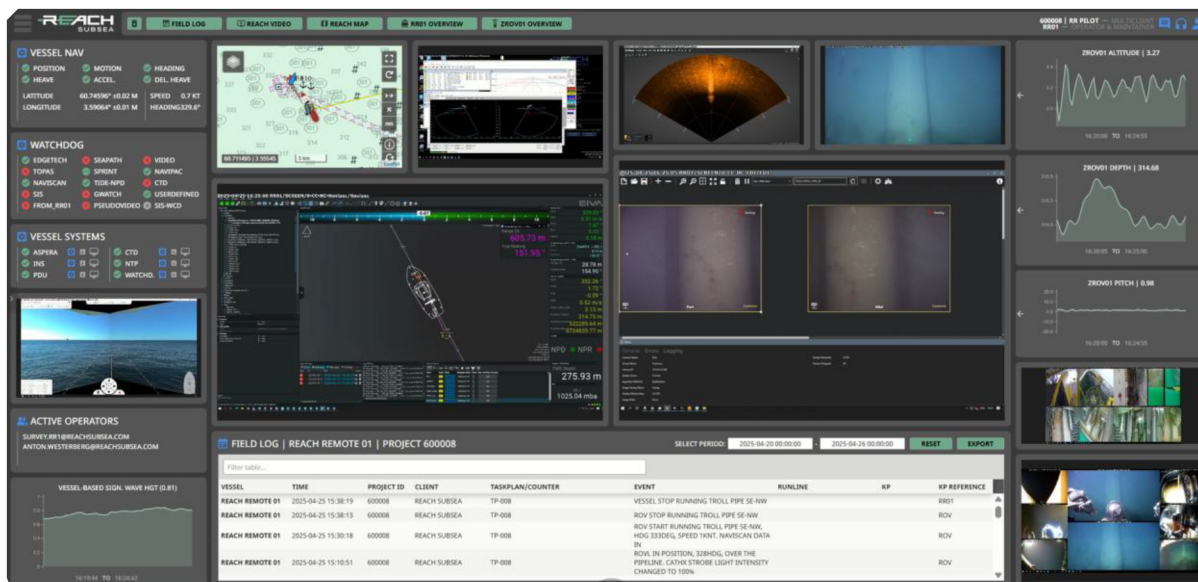


Figure 4—Reach Horizon Example during pipeline inspection [Image courtesy of Reach Subsea]

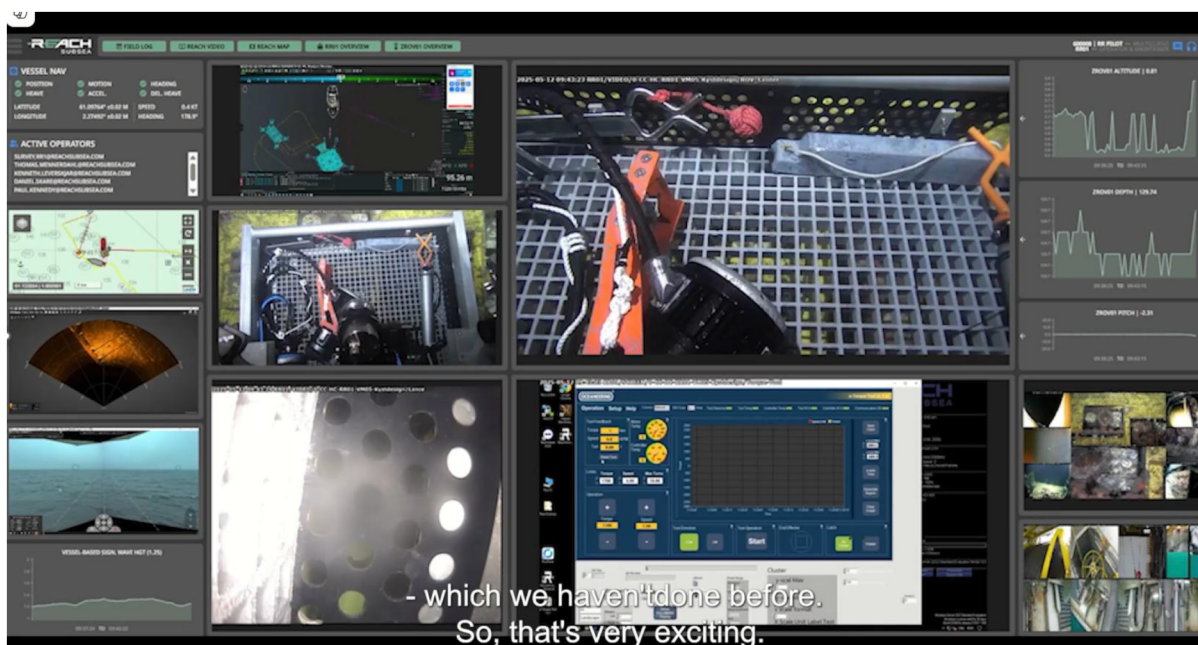


Figure 5—Reach Horizon Example during subsea intervention and valve operations [Image courtesy of Reach Subsea]

Following completion of the JIP, Anders Opedal, President and CEO Equinor said:

"The Reach Remote 1 is a 24-meter-long uncrewed vessel, operated from a control room onshore. This is an example to me of how an idea - a "what if" - goes from being a thought and a vision, into reality. This takes perseverance, willingness to take risks, trust and collaboration to achieve.

So far, Reach Remote 1 has demonstrated pipeline inspection, seabed mapping, gravimetric survey (including extensive use of manipulator), structural inspection, subsea valve operations, Class IV electric torque tool tests and photogrammetry. The vessel recently performed what we believe is the world's first unmanned IMR-operation at Gullfaks".

Challenges and Lessons Learned

While the technology has proven robust, several challenges were encountered:

- USVs come in many shapes and sizes, and our understanding of the application of these different solutions in the marketplace can, at times be immature.
- Adapting regulatory frameworks to accommodate uncrewed offshore operations.
 - Having a USV operate in one country is a challenge, to meet regulations and gain approvals
 - Once you consider how to operate a USV across international boundaries with onshore control centres in one location the scenarios become exponentially more complex.
- Ensuring cybersecurity across a distributed control network is critical. The vessel at sea, control infrastructure onshore and all terminal applications become mission critical hardware and robust processes are required to ensure full compliance and security.
- Training and certifying onshore operators and establishing new workflows for remote management.

These lessons have guided ongoing refinements to the Reach Remote platform, in close collaboration with regulatory authorities and industry partners.

The Regulatory Environment for USV Operations

The regulatory environment for Uncrewed Surface Vessels (USVs) is still evolving and presents one of the most significant barriers to large-scale adoption, and commercialisation. Existing maritime rules, such as those under the International Maritime Organization (IMO) and national maritime authorities, were primarily written for conventional, crewed ships. As a result, applying these frameworks to USVs raises several challenges.

- **International Conventions:** Key instruments such as the International Regulations for Preventing Collisions at Sea (COLREGs) and the Safety of Life at Sea (SOLAS) Convention were designed with human watchkeepers and onboard crews in mind. Translating these obligations to a remotely operated or autonomous system introduces ambiguity. For example, how to interpret "maintaining a proper lookout" when sensors, cameras, and remote operators replace a physical bridge crew remains a subject of debate.
- **National and Regional Regulations:** Different flag states and coastal states are developing their own policies for USV operations. Some, like Norway and the UK, have introduced pilot schemes and exemptions that enable trials in designated areas. Others remain cautious, requiring crewed "assisting vessels" or imposing strict permitting processes. This patchwork approach complicates global deployment, particularly for companies seeking to operate USVs across multiple jurisdictions in the same sense as many operate traditional crewed vessels.
- **Certification and Classification:** Classification societies such as DNV, Lloyd's Register, Bureau Veritas and ABS are issuing guidelines for autonomous and remotely operated vessels. While these frameworks provide an important pathway to safety assurance, they are not yet harmonised. Variability in standards creates uncertainty for operators and potentially delays widespread commercial adoption.
- **Liability and Insurance:** Questions of liability in the event of collision, grounding, or pollution are unresolved. Determining responsibility—whether it lies with the remote operator, the system developer, the vessel owner, or the flag state—remains legally complex. Insurers have been cautious, often requiring additional risk assessments or premiums.
- **Operational Restrictions:** Until regulations catch up, many (though not all) USVs are limited to trials, nearshore operations, or controlled corridors. This restricts their ability to demonstrate full

value in offshore and transoceanic operations, where their endurance and cost advantages may be greatest.

The lack of harmonised international regulation risks slowing down the sector's growth. Companies must navigate a patchwork of approvals, limiting efficiency and scalability. Without a consistent global framework, operations may remain fragmented, with innovation concentrated in regions more open to experimentation, with a perceived different attitude to risk of USV operations.

Future Outlook

The trajectory of USVs within the maritime sector is clear; expansion across multiple sectors, including defence, offshore energy, and marine research. The scale and capability of USVs will evolve. Some smaller USVs will become more capable and the larger USVs will become more complex and able to deliver significant payload advantages. From a Reach Subsea perspective, we anticipate the deployment of a globally distributed fleet of uncrewed vessels, each capable of providing 24/7 subsea intervention, surveillance, and survey. Shaping a safer, more efficient, and environmentally responsible industry.

Novelty and Industry Impact

Reach Remote is the world's first truly uncrewed surface vessel capable of deploying a fully electric WROV for subsea operations. The platform is designed and manufactured by Kongsberg, incorporating pioneering innovations in autonomous navigation, energy management, and maritime robotics. Its ability to operate over-the-horizon, without crew at sea, sets a new standard for what is possible in subsea operations.

The implications are profound:

- Substantial reduction in offshore personnel exposure and associated risks.
- Transformational cost savings in energy, defence, and survey operations.
- Enhanced environmental performance, supporting net-zero and ESG objectives.
- Rapid scalability and deployment, enabled by digitalisation and remote access.

Conclusion

Reach Remote demonstrates the art of the possible: a future where subsea operations are safer, more sustainable, and radically more efficient. By leveraging advanced uncrewed surface vessels and electric WROV technology, the industry can realise unprecedented operational flexibility and resilience. Our experience to date—spanning real-world deployments, cross-industry collaboration, and regulatory engagement—underscores the need for continued innovation and adoption of uncrewed maritime solutions.

As we look ahead, Reach Remote stands ready to serve as the vanguard of this new era, delivering operational excellence and environmental stewardship in subsea work across the globe. It's a true example of what is possible for USVs in the maritime sector.